

Project Testing

Objective

The objective of performance testing is to evaluate the accuracy, efficiency, and reliability of the developed Credit Card Approval Prediction System. The application was tested using different machine learning algorithms and validated through the Flask web interface to ensure that prediction results are generated correctly and within a short response time.

Test Case	Expected	Result
Load Home Page	Page loads	Pass
Enter Valid Input	Accepted	Pass
Predict Transaction	Result shown	Pass
Invalid Input	Error handled	Pass

3. Model Performance

My metrics:

- Logistic Regression
- Decision Tree
- Random Forest

	Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
0	Logistic Regression	0.9990	0.7000	0.6632	0.6811	0.8313
1	Decision Tree	0.9990	0.6907	0.7053	0.6979	0.8524
2	Random Forest	0.9995	0.9718	0.7263	0.8313	0.8631

4. Conclusion

The Random Forest Classifier achieved the highest overall performance among all evaluated models and was selected as the final prediction model due to its superior accuracy and balanced performance across evaluation metrics.